KS3 DIGITAL TECHNOLOGY

SPACE TECHNOLOGY AND SATELLITES

The computing above our heads: how satellites stay in orbit, how GPS pinpoints your location, how Earth observation and AI monitor our planet, how satellite internet works, why space debris is a growing problem, and how to design your own CubeSat mission.

LESSONS

1 Satellites and orbits

2 Satellite navigation

3 Watching the Earth

4 Space internet and communications

5 Space junk

6 Mission design and the space industry

LESSON 1 OF 6

SATELLITES AND ORBITS

A satellite stays in orbit because it travels sideways fast enough that, as gravity pulls it down, the Earth curves away beneath it.

STARTER · 5 MINUTES

DO NOW

If you throw a ball sideways it falls to the ground. What would happen if you could throw it at 28,000 km/h?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Explain how satellites stay in orbit, compare types of orbit and describe the parts of a satellite

SUCCESS CRITERIA: I CAN...

- ✓ I can explain how a satellite stays in orbit
- ✓ I can compare LEO, MEO and GEO orbits and their uses
- ✓ I can describe the main parts of a satellite

FALLING AROUND THE EARTH

- A satellite is an object that orbits a larger object
- Gravity pulls it towards Earth all the time
- It moves sideways so fast (about 28,000 km/h in low orbit) that it keeps missing the ground
- It's "falling around" the Earth

KEY WORDS

SATELLITE	An object that orbits a larger object; artificial satellites are machines launched into orbit.
ORBIT	The curved path an object follows around a planet or star.
LOW EARTH ORBIT	An orbit 160-2,000 km above Earth, used by the ISS and many satellites.
GEOSTATIONARY ORBIT	An orbit 35,786 km above the equator where a satellite stays over the same spot.
PAYLOAD	The instruments a satellite carries to do its job.

TYPES OF ORBIT

- LEO (low Earth orbit): 160-2,000 km; ISS, Starlink, imaging satellites
- MEO (medium Earth orbit): about 20,000 km; GPS and Galileo
- GEO (geostationary orbit): 35,786 km; stays over one spot; TV and weather
- Higher orbits take longer to go round

ACTIVITY · 15 MINUTES · PAIRS

ORBIT MATCH

1. Match the eight satellite cards (e.g. a TV satellite, the ISS, a GPS satellite, a weather satellite watching Europe, a Starlink satellite) to the right orbit.

2. For each, explain why that orbit suits its job.

3. Work out how many times the ISS orbits the Earth in a day (it takes about 90 minutes per orbit).

In your workbook, answer:

? Why do TV satellites use geostationary orbit?

? How many times does the ISS orbit Earth in a day?

Your workbook: Lesson 1 · tick each step off as you go

INSIDE A SATELLITE

- Power: solar panels and batteries
- Computer (on-board data handling) to control everything
- Communications: antennas to talk to ground stations
- Payload: the instruments that do the job, e.g. cameras or radios

QUICK CHECK

1. Why doesn't a satellite fall to Earth?

A It moves sideways so fast the Earth curves away beneath it
B There's no gravity in space
C It has wings
D It's held up by air

2. Which orbit stays above the same point on Earth?

A Medium Earth orbit
B No orbit
C Geostationary orbit
D Low Earth orbit

3. Which orbit is the ISS in?

A It doesn't orbit
B Geostationary orbit
C Medium Earth orbit
D Low Earth orbit

4. What is a satellite's payload?

A Its price
B The instruments that do its job
C Its launch rocket
D Its fuel

QUICK CHECK: ANSWERS

1. **A: It moves sideways so fast the Earth curves away beneath it**
   It's falling around the Earth.

2. **C: Geostationary orbit**
   It matches Earth's spin.

3. **D: Low Earth orbit**
   About 400 km up.

4. **B: The instruments that do its job**
   e.g. cameras, radios.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain why a satellite stays in orbit.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain how a satellite stays in orbit
- ✓ I can compare LEO, MEO and GEO orbits and their uses
- ✓ I can describe the main parts of a satellite

LESSON 2 OF 6

SATELLITE NAVIGATION

Satellite navigation (GNSS) uses satellites that broadcast precise time signals from atomic clocks.

STARTER · 5 MINUTES

DO NOW

How does your phone know where you are, even when you're not connected to Wi-Fi?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Explain how satellite navigation systems work out a receiver's position, and describe their uses and limitations

SUCCESS CRITERIA: I CAN...

- ✓ I can describe GNSS systems such as GPS and Galileo
- ✓ I can explain how trilateration works out a position
- ✓ I can explain why accurate timing matters and describe limitations

SATELLITE NAVIGATION

- GNSS: Global Navigation Satellite System
- GPS (USA), Galileo (Europe), GLONASS (Russia), BeiDou (China)
- About 24-30 satellites per system, about 20,000 km up
- Your phone often uses several systems together

KEY WORDS

GNSS	Global Navigation Satellite System: satellites that let receivers work out their position.
GPS	The Global Positioning System, the USA's satellite navigation system.
TRILATERATION	Working out a position from distances to several known points.
ATOMIC CLOCK	An extremely accurate clock used in navigation satellites.

HOW IT WORKS

- Each satellite broadcasts its position and the exact time
- Your receiver measures how long each signal took to arrive
- Distance = speed of light × time
- With 4 satellites, it calculates your position (trilateration) and corrects its own clock

ACTIVITY · 25 MINUTES · GROUPS OF 4

HUMAN GPS

1. Three pupils stand in the playground or hall as "satellites", each holding a rope or tape measure.

2. A fourth pupil (the "receiver") is placed secretly. Each satellite measures the distance to the receiver.

3. On a scale map, draw circles of the right radius around each satellite's position and find where they cross.

4. Repeat with one measurement deliberately 1 m wrong and see what happens.

In your workbook, answer:

? How many circles were needed to find the receiver's position on a flat map, and why?

Your workbook: Lesson 2 · tick each step off as you go

? What happened when one measurement was wrong, and what does this show?

TIMING IS EVERYTHING

- Radio signals travel at about 300,000 km per second
- An error of one millionth of a second = about 300 m wrong
- So satellites carry ultra-precise atomic clocks
- Banks, power grids and phone networks also rely on GNSS timing

QUICK CHECK

1. What does GNSS stand for?

A Global Network Signal System
B Ground Navigation Station System
C Great New Satellite Service
D Global Navigation Satellite System

2. Which navigation system is run by Europe?

A GLONASS
B Galileo
C GPS
D BeiDou

3. What is the minimum number of satellites a receiver needs for a full position fix?

A 4
B 2
C 1
D 10

4. Why do navigation satellites carry atomic clocks?

A To save power
B To tell astronauts the time
C Tiny timing errors cause big position errors
D For decoration

QUICK CHECK: ANSWERS

1. **D: Global Navigation Satellite System**
GPS is one example.

2. **B: Galileo**
Europe's GNSS.

3. **A: 4**
Three for position, one for clock correction.

4. **C: Tiny timing errors cause big position errors**
Microseconds matter.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain how a GNSS receiver works out its position.

Look back at today's success criteria: can you tick them all?

- ✓ I can describe GNSS systems such as GPS and Galileo
- ✓ I can explain how trilateration works out a position
- ✓ I can explain why accurate timing matters and describe limitations

LESSON 3 OF 6

WATCHING THE EARTH

Earth observation satellites use cameras and sensors, including radar that can see through cloud, to monitor weather, climate, farming, floods and wildfires.

STARTER · 5 MINUTES

DO NOW

Name something that is happening on Earth right now that you could spot from space.

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Describe how Earth observation satellites and AI are used to monitor the planet and respond to disasters

SUCCESS CRITERIA: I CAN...

- ✓ I can describe the sensors Earth observation satellites use
- ✓ I can describe uses of Earth observation data
- ✓ I can explain how AI helps analyse satellite images

EYES IN THE SKY

- Earth observation: studying Earth from space
- Optical cameras: like a phone camera, but need daylight and clear skies
- Infrared: heat from fires, temperature of the sea
- Radar: sees through cloud and works at night

KEY WORDS

EARTH OBSERVATION	Gathering information about the Earth from space.
REMOTE SENSING	Measuring something from a distance without touching it.
RESOLUTION	The size of the smallest detail an image can show.
COPERNICUS	The European Union's Earth observation programme, which provides free satellite data.
SYNTHETIC APERTURE RADAR	A satellite radar that can image the Earth through cloud and at night.

WHAT WE WATCH

- Weather forecasting and climate change: ice, sea level, temperature
- Farming: crop health, drought
- Disasters: floods, wildfires, earthquakes
- Pollution, deforestation, illegal fishing

AI AND BIG DATA

- Satellites send back huge amounts of data every day
- AI spots changes: new buildings, flooded fields, burnt forest
- Copernicus (EU) data is free for anyone
- Privacy: how much should satellites be able to see?

ACTIVITY · 25 MINUTES · GROUPS OF 3

DISASTER RESPONSE TEAM

1. Your teacher gives you "before" and "after" satellite images of a flooded area (or use a free online viewer such as Copernicus Browser).
2. Identify the areas that have flooded and the roads that are cut off.
3. Decide which sensor would work best if it's still cloudy, and why.
4. Write a short briefing for the rescue team: where to go first and why.

In your workbook, answer:

? Which sensor would work best on a cloudy day, and why?

? Write your rescue briefing.

Your workbook: Lesson 3 · tick each step off as you go

QUICK CHECK

1. What is Earth observation?

A Gathering information about Earth from space
B Studying rocks by hand
C Looking at stars
D Watching TV

2. Which sensor can see through cloud?

A Optical camera
B Human eyes
C A microphone
D Radar

3. What is image resolution?

A The satellite's height
B The file name
C The size of the smallest detail an image can show
D The colour of the image

4. What is Copernicus?

A A planet
B A GPS system
C The EU's Earth observation programme with free data
D A rocket

QUICK CHECK: ANSWERS

1. **A: Gathering information about Earth from space**
Satellites monitor Earth.

2. **D: Radar**
Radar works in all weather.

3. **C: The size of the smallest detail an image can show**
Smaller = more detail.

4. **C: The EU's Earth observation programme with free data**
Sentinel satellites.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Describe three uses of Earth observation satellites.

Look back at today's success criteria: can you tick them all?

- ✓ I can describe the sensors Earth observation satellites use
- ✓ I can describe uses of Earth observation data
- ✓ I can explain how AI helps analyse satellite images

LESSON 4 OF 6

SPACE INTERNET AND COMMUNICATIONS

Communication satellites relay signals between distant places.

STARTER · 5 MINUTES

DO NOW

How could you get fast internet on a remote island, a ship at sea or a farm on a mountain?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Describe how satellites provide communications and internet access, and evaluate mega-constellations

SUCCESS CRITERIA: I CAN...

- ✓ I can explain how communication satellites relay signals
- ✓ I can explain latency and why LEO satellites reduce it
- ✓ I can evaluate the benefits and concerns of mega-constellations

RELAYING SIGNALS

- Communication satellites receive a signal and send it on
- Ground stations connect satellites to the internet
- Used for TV, phone calls, ships, aircraft and remote areas
- Deep space missions talk to Earth through giant dish antennas

KEY WORDS

COMMUNICATION SATELLITE	A satellite that relays radio signals between places on Earth.
LATENCY	The delay between sending data and it arriving.
MEGA-CONSTELLATION	A group of hundreds or thousands of satellites working together.
GROUND STATION	A site on Earth with antennas for communicating with satellites.

THE LATENCY PROBLEM

- Latency: the delay before data arrives
- GEO satellites are 35,786 km away: a round trip takes over half a second
- LEO satellites are about 550 km away: much lower latency
- But LEO satellites move fast, so you need thousands to cover the Earth

MEGA-CONSTELLATIONS

- Starlink: thousands of satellites, more than half of all active satellites
- Other companies and countries are building their own
- Satellites can now connect directly to ordinary phones in some places
- Concerns: astronomy, debris, and control by a few companies

ACTIVITY · 20 MINUTES · PAIRS

CONNECT THE ISLAND

1. A small island off the coast of Donegal needs internet for its school, a fish farm and holiday cottages.
2. Compare three options using the fact cards: an undersea cable, a geostationary satellite service and a LEO satellite service.
3. Score each option for speed, latency, cost and reliability.
4. Recommend an option and calculate the latency for the geostationary option.

In your workbook, answer:

? Which option do you recommend, and why?

? Why does the geostationary option have high latency?

Your workbook: Lesson 4 · tick each step off as you go

QUICK CHECK

1. What does a communication satellite do?

A Launches rockets
B Takes photos only
C Measures temperature
D Receives signals and sends them on to another place

2. What is latency?

A The delay before data arrives
B Signal strength
C The amount of data
D The satellite's weight

3. Why do LEO satellites have lower latency than GEO?

A They're much closer to Earth
B They use cables
C They're bigger
D They're older

4. What is a mega-constellation?

A A ground station
B Hundreds or thousands of satellites working together
C A huge star
D One giant satellite

QUICK CHECK: ANSWERS

1. **D: Receives signals and sends them on to another place**
It relays signals.

2. **A: The delay before data arrives**
Time delay.

3. **A: They're much closer to Earth**
Shorter distance.

4. **B: Hundreds or thousands of satellites working together**
e.g. Starlink.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain why thousands of satellites are needed for LEO internet coverage.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain how communication satellites relay signals
- ✓ I can explain latency and why LEO satellites reduce it
- ✓ I can evaluate the benefits and concerns of mega-constellations

LESSON 5 OF 6

SPACE JUNK

Space debris includes dead satellites, used rocket stages and fragments from collisions and explosions.

STARTER · 5 MINUTES

DO NOW

A paint fleck hits a window at 28,000 km/h. What damage could it do?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Explain the causes and dangers of space debris and evaluate ways to make space sustainable

SUCCESS CRITERIA: I CAN...

- ✓ I can describe what space debris is and where it comes from
- ✓ I can explain why debris is dangerous, including the Kessler syndrome
- ✓ I can evaluate solutions to the space debris problem

WHAT IS SPACE JUNK?

- Dead satellites and used rocket stages
- Fragments from explosions, collisions and anti-satellite tests
- Tens of thousands of tracked objects; perhaps over a million pieces larger than 1 cm
- At orbital speed, even a paint fleck can crack a window

KEY WORDS

SPACE DEBRIS	Human-made objects in orbit that no longer do a useful job.
KESSLER SYNDROME	A possible chain reaction of collisions that creates more and more debris.
DEORBIT	To bring a satellite down so it burns up in the atmosphere.
COLLISION AVOIDANCE	Moving a satellite to avoid hitting debris or another satellite.

THE KESSLER SYNDROME

- One collision creates thousands of fragments
- Those fragments cause more collisions
- A chain reaction could make some orbits unusable for generations
- We rely on satellites for navigation, communication and weather

CLEANING UP SPACE

- Tracking debris and moving satellites to avoid collisions
- Design satellites to deorbit and burn up at the end of their mission
- Active debris removal: missions to capture junk (UK companies are leading tests)
- International rules and responsible behaviour

ACTIVITY · 25 MINUTES · GROUPS OF 3

JUNK BUSTERS

1. Read the four debris-removal ideas: magnetic docking, a giant net, a robotic arm, and a laser that nudges junk down.
2. Score each idea for cost, risk, how many objects it could remove and whether it could be misused.
3. Choose the best idea, or combine ideas.
4. Write three "rules of the road" all satellite operators should follow.

In your workbook, answer:

? Which idea did you choose, and why?

? Write your three rules of the road.

Your workbook: Lesson 5 · tick each step off as you go

QUICK CHECK

1. What is space debris?

A Stars
B Moon rocks
C Asteroids
D Human-made objects in orbit that no longer do a useful job

2. Why is small debris dangerous?

A It's radioactive
B It travels so fast it hits with huge energy
C It isn't
D It's magnetic

3. What is the Kessler syndrome?

A A chain reaction of collisions creating more and more debris
B A space illness
C A rocket engine
D A type of orbit

4. What does it mean to deorbit a satellite?

A Switch it on
B Bring it down so it burns up in the atmosphere
C Paint it
D Launch it higher

QUICK CHECK: ANSWERS

1. **D: Human-made objects in orbit that no longer do a useful job**
Space junk.

2. **B: It travels so fast it hits with huge energy**
Speed makes it deadly.

3. **A: A chain reaction of collisions creating more and more debris**
Collisions cascade.

4. **B: Bring it down so it burns up in the atmosphere**
Clears orbit.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain how space debris could make an orbit unusable.

Look back at today's success criteria: can you tick them all?

- ✓ I can describe what space debris is and where it comes from
- ✓ I can explain why debris is dangerous, including the Kessler syndrome
- ✓ I can evaluate solutions to the space debris problem

LESSON 6 OF 6

MISSION DESIGN AND THE SPACE INDUSTRY

CubeSats are small, low-cost satellites built from 10 cm cubes that let universities, schools and start-ups reach space.

STARTER · 5 MINUTES

DO NOW

If you could put one sensor in space to help your local area, what would it measure?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Plan a small satellite mission and describe careers and opportunities in the space industry

SUCCESS CRITERIA: I CAN...

- ✓ I can describe what a CubeSat is
- ✓ I can plan a satellite mission including payload, orbit, power and communications
- ✓ I can describe careers and opportunities in the space industry

CUBESATS

- Small satellites built from 10 cm cube units (1U); common sizes 1U, 3U, 6U
- Use off-the-shelf parts like phone components
- Cheap enough for universities, schools and start-ups
- Share a ride into space with other satellites

KEY WORDS

CUBESAT	A small satellite built from standard 10 cm cube units.
MISSION OBJECTIVE	The goal a space mission is designed to achieve.
GROUND SEGMENT	The ground stations, software and people that control a satellite and process its data.
SPACEPORT	A site for launching rockets or spacecraft.

PLANNING A MISSION

- Objective: what problem will it solve?
- Payload: what instruments are needed?
- Orbit, power, communications and data
- End of life: how will it deorbit safely?

THE UK IN SPACE

- UK companies build many of the world's small satellites (Glasgow builds more than almost anywhere outside the US)
- UK spaceports such as SaxaVord in Shetland aim to launch rockets from the UK
- Northern Ireland firms work in space software, data and components
- Careers: software, data science, engineering, law, science, communications

ACTIVITY · 35 MINUTES · GROUPS OF 3

CUBESAT CHALLENGE

1. Choose a mission to help Northern Ireland: water quality in Lough Erne, flood warnings, peatland health, or farm crop monitoring.
2. Complete the mission planning checklist.
3. Design a mission patch and name your satellite.
4. Pitch your mission in 2 minutes, including how it will avoid becoming space junk.

In your workbook, answer:

? Complete your mission plan: objective, payload and orbit.

? How will your CubeSat avoid becoming space junk?

Your workbook: Lesson 6 · tick each step off as you go

QUICK CHECK

1. What is a CubeSat?

A A space station
B A type of rocket
C A small satellite built from 10 cm cube units
D A cube-shaped planet

2. Why are CubeSats cheaper than traditional satellites?

A They don't need power
B They're made of cardboard
C They never leave Earth
D Standard sizes, off-the-shelf parts and shared launches

3. What is the ground segment of a mission?

A Soil samples
B The ground stations, software and people that control the satellite and process data
C The launch pad only
D The bottom of the rocket

4. Which is a space industry job?

A Lamplighter
B Town crier
C Satellite data scientist
D Typewriter repairer

QUICK CHECK: ANSWERS

1. **C: A small satellite built from 10 cm cube units**
Low-cost satellites.

2. **D: Standard sizes, off-the-shelf parts and shared launches**
Standardisation lowers cost.

3. **B: The ground stations, software and people that control the satellite and process data**
Earth-based part of the mission.

4. **C: Satellite data scientist**
Most space jobs are on Earth.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Describe three things that must be planned for a CubeSat mission.

Look back at today's success criteria: can you tick them all?

- ✓ I can describe what a CubeSat is
- ✓ I can plan a satellite mission including payload, orbit, power and communications
- ✓ I can describe careers and opportunities in the space industry

END OF TOPIC

SPACE TECHNOLOGY AND SATELLITES

Revise each lesson in your workbook: the notes, key words and quick checks.
Then take the topic test when Mr Butler sets it.